Srujay Reddy Vangoor

AI Software Engineer | Backend Systems | Production LLM Platforms | Cloud Infrastructure

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Software Engineer with 6+ years of experience delivering production AI systems, cloud native backend services, and full stack applications. Experienced in retrieval augmented generation, agentic workflows, LLM evaluation, knowledge graph augmented retrieval, and distributed microservice architectures. Skilled in deploying systems on AWS and Azure, building observability and reliability into production platforms, and modernizing legacy applications into secure cloud native solutions.

TECHNICAL SKILLS

AI & LLM Systems: RAG, Agentic Workflows, MCP, LangChain, LangGraph, CrewAI, Vector Search (FAISS, Pinecone), LLM Evaluation (RAGAS, LangSmith), Prompt Engineering, Hugging Face Transformers, PyTorch, Scikit learn

Backend & APIs: Python (FastAPI, Django), C# (ASP.NET Core), Node.js (Express), REST APIs, GraphQL, WebSockets

Cloud: AWS, Azure, Docker, Kubernetes, Terraform, CI/CD, OpenTelemetry, Grafana, Prometheus, Loki, Tempo

Data Systems: PostgreSQL, Redis, Kafka, DynamoDB, Microsoft SQL Server, Snowflake, Power BI

Frontend & Mobile: React, Next.js, TypeScript, JavaScript, React Native

EXPERIENCE

AI Software Engineer
Feb 2025 – Present

California Department of Developmental Services (DDS) — Sacramento, CA (Hybrid)

- Built a conversational RAG system from scratch for internal report analysis, implementing retrieval evaluation pipelines to measure answer grounding, retrieval relevance, and response quality.
- Engineered an agentic AI workflow orchestration platform using CrewAI with deterministic execution, schema validation, and persistent memory, reducing multi step failures from 23% to under 7% in production.
- Improved retrieval quality by increasing Hit@1 from 0.35 to 0.63 through a knowledge graph augmented retrieval layer with subgraph expansion, hybrid embeddings, and relevance pruning.
- Implemented an enterprise AI observability platform for Claude Code using OpenTelemetry and Grafana Alloy, centralizing metrics, logs, and traces in Prometheus, Loki, Tempo, and Grafana for engineering observability.
- Designed and deployed AWS based FastAPI microservices supporting more than 200K API requests per day with sub second P99 latency for secure healthcare workflows.
- Built a full stack AI workflow intake portal using Next.js, TypeScript, Express, AWS, and Terraform, reducing submission time from 30 to 10 minutes through automated validation and review workflows.

Software Engineer
Jan 2024 – Feb 2025

California Department of Conservation (DOC) — Sacramento, CA (Remote)

- Engineered a seismic modeling service on AWS, integrating Python and OpenQuake with legacy PHP and Perl systems to deliver ground motion predictions in under three seconds per request.
- Reduced data pipeline latency by optimizing Kafka ingestion of more than 10 GB per day from earthquake monitoring stations, enabling near real time analytics with Apache Druid.
- Implemented identity and access management controls including RBAC, MFA, AWS Secrets Manager, and Azure Entra ID SSO to secure access across more than 60 monitoring stations.
- Improved geospatial application responsiveness and reduced recurring downtime by optimizing Leaflet rendering, GeoServer integration, and performance bottlenecks affecting large datasets.

Software Application Developer Intern
Jan 2023 – Jun 2024

Population Research Center (PRC), Sacramento State — Sacramento, CA

- Built a staff management system using C#, Entity Framework, and SQL Server, optimizing stored procedures and indexing to reduce report generation and data retrieval times by 40%.
- Designed and automated health data ETL pipelines using .NET and SQL Server, migrating Access and Oracle datasets into centralized systems while supporting HIPAA aligned data handling practices.

- Developed React based data entry tools and analytics dashboards for statewide public health studies, improving data collection quality through automated validation and quality checks.
- Implemented role based access controls and server side validation across internal .NET applications to protect sensitive public health and research data.

Software Engineer
Jan 2021 – Aug 2022

Human Sciences Research Group (HSRG), IIIT Hyderabad
Hyderabad, India

- Architected a cloud native archival and analytics platform for social media and news streams, enabling near real time analysis of more than 1 million social media posts and articles during the Indian Farmers' Protests.
- Improved sentiment classification F1 score by 8% over baseline by fine tuning BERT and IndicBERT models using Hugging Face Transformers for multilingual social media analysis.
- Built 13 secure REST API endpoints using C# and ASP.NET Core for high volume data ingestion and metadata management, and developed React dashboards for exploratory analysis and reporting.

Software Engineer
Jul 2018 – Dec 2020

Change and Continuity in Spiti Valley, ICSSR Sponsored Project
Hyderabad, India

- Built a GIS platform using Python, Django, PostgreSQL, and PostGIS to map more than 280 heritage sites across 25 villages, enabling spatial search, analytics, and interactive visualization.
- Developed responsive React and offline capable React Native applications using Leaflet, reducing page load times by 60% in low bandwidth field environments.
- Automated Python ETL and spatial processing workflows integrating surveys, census, and archival datasets into a unified geospatial repository with validation and incremental updates.

PROJECTS

AI Software Engineering Memory – Repository Knowledge Standard

- Engineered and open sourced a lightweight repository memory standard that enables AI coding agents to retain architectural context, engineering decisions, and repository conventions across development sessions.
- Designed reusable planning, review, and documentation templates and validated the framework across multiple repositories to standardize AI assisted development and accelerate repository onboarding.

ContextServer – MCP Tool Infrastructure for AI Agents

- Built an MCP platform enabling AI agents to inspect and reason over enterprise codebases through secure filesystem access, Git history analysis, and semantic documentation retrieval.
- Architected an extensible framework with observability, execution tracing, and dynamic capability discovery, allowing new agent capabilities through declarative tool registration without changing core routing logic.

CA DMV RAG System

- Developed a RAG system using FastAPI, FAISS, and Sentence Transformers, supporting streaming responses, citation grounded answers, semantic reranking, retrieval across multiple documents, and confidence scoring.
- Implemented evaluation and observability pipelines with distributed tracing, metrics, automated testing, and CI/CD to measure retrieval relevance, citation grounding, and end to end system performance.

VeryFastChat – Anonymous Real Time Chat Platform

- Engineered a cloud native real time messaging platform using Next.js, FastAPI, and Supabase, enabling instant room creation, persistent chat history, shareable invitations, and moderated multiuser collaboration.
- Established production readiness through Redis rate limiting, configuration validation, distributed tracing, health checks, automated CI/CD pipelines, and secure deployments on Vercel and Render.

EDUCATION

Master of Science (MS) in Computer Science
Aug 2022 – Jan 2025

California State University, Sacramento
Sacramento, California

Bachelor of Technology (BTech) in Computer Science
Aug 2015 – Jul 2019

International Institute of Information Technology, Hyderabad (IIIT-Hyderabad)
Hyderabad, India